

COMPSCI 752

Big Data Management

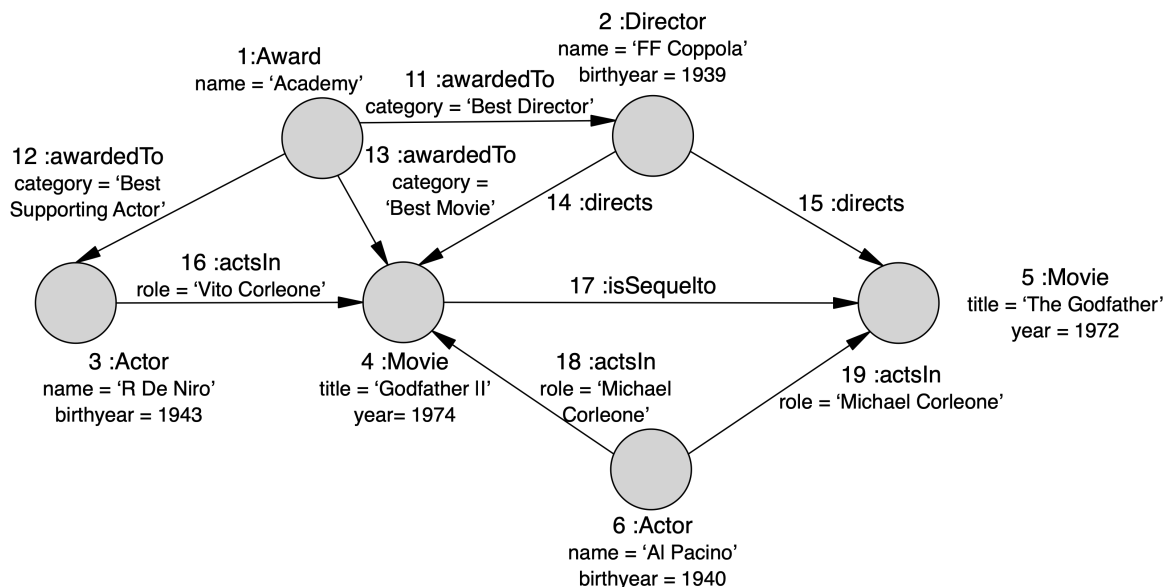
Strategic Exercise 4

Graph Models - Model Solutions

Exercise 1 - Property graphs.

Imagine the scenario where we would like to develop a graph database for movie data. The minimum requirements are to model actors, directors, movies, sequels of movies, awards, their properties, and their relationships.

- Write down a small property graph with real-world data.
- Explain the properties and relationships on examples of your graph.



Actors have a *name* and a *birthyear*, such as Robert de Niro born in 1943. Similarly, directors have a *name* and a *birthyear*, such as FF Coppola born in 1939. Movies have a *title* and a *year* they were produced in, such as The Godfather produced in 1972. Awards have a *name*, such the Academy awards. Actor may act in movies, such as the actor R De Niro acting in the role of Vito Corleone in the movie Godfather II. Directors direct movies, such as FF Coppola directing the movie The Godfather. Awards can be awarded to different entities, such as the Academy award being awarded to the Actor R De Niro in the category Best Supporting Actor. Finally, a movie can be a sequel to another movie, such as the movie Godfather II being a sequel to movie The Godfather.

Exercise 2 - Formal model.

For the property graph from the first Exercise,

- a. write down the formal definition of a property graph in terms of $G = (V, E, \eta, \lambda, \nu)$.

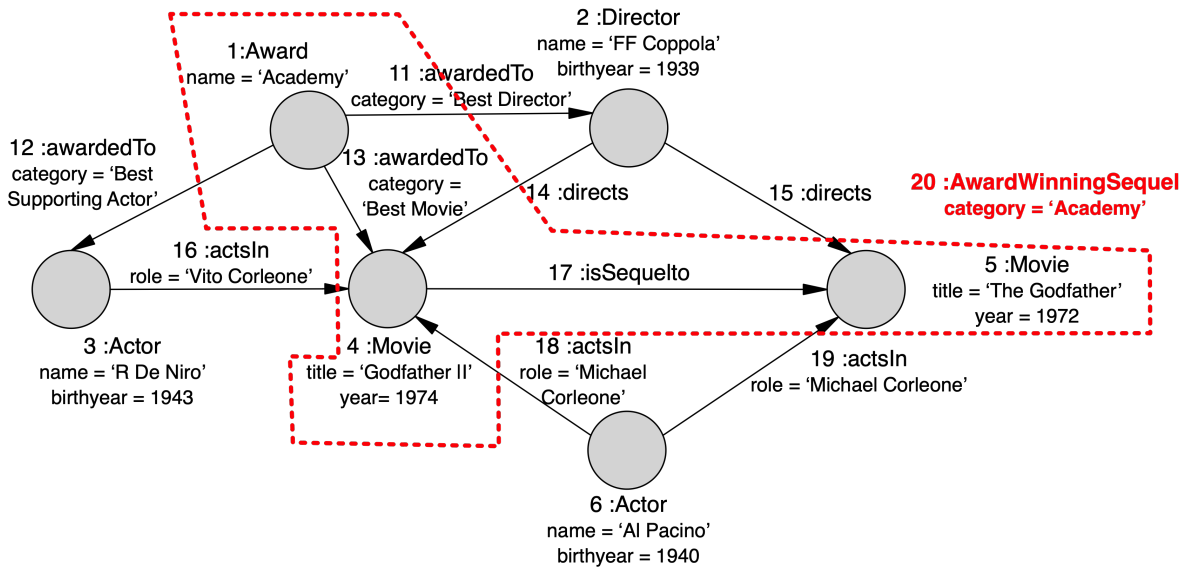
The property graph that constitutes one solution for the first exercise is formally defined as follows:

- $V = \{1, 2, 3, 4, 5, 6\}$
- $E = \{11, 12, 13, 14, 15, 16, 17, 18, 19\}$
- η :
 - $11 \mapsto (1, 2)$
 - $12 \mapsto (1, 3)$
 - $13 \mapsto (1, 4)$
 - $14 \mapsto (2, 4)$
 - $15 \mapsto (2, 5)$
 - $16 \mapsto (3, 4)$
 - $17 \mapsto (4, 5)$
 - $18 \mapsto (6, 4)$
 - $19 \mapsto (6, 5)$
- λ :
 - $1 \mapsto \text{:Award}$
 - $2 \mapsto \text{:Director}$
 - $3, 6 \mapsto \text{:Actor}$
 - $4, 5 \mapsto \text{:Movie}$
 - $11, 12, 13 \mapsto \text{:awardedTo}$
 - $14, 15 \mapsto \text{:directs}$
 - $16, 18, 19 \mapsto \text{:actsIn}$
 - $17 \mapsto \text{:isSequelTo}$
- ν :
 - $(1, \text{name}) \mapsto \text{'Academy'}$
 - $(2, \text{name}) \mapsto \text{'FF Coppola'}$
 - $(2, \text{birthyear}) \mapsto 1939$
 - $(3, \text{name}) \mapsto \text{'R De Niro'}$
 - $(3, \text{birthyear}) \mapsto 1943$
 - $(4, \text{title}) \mapsto \text{'Godfather II'}$
 - $(4, \text{year}) \mapsto 1974$
 - $(5, \text{title}) \mapsto \text{'The Godfather'}$
 - $(5, \text{year}) \mapsto 1972$
 - $(6, \text{name}) \mapsto \text{'Al Pacino'}$
 - $(6, \text{birthyear}) \mapsto 1940$
 - $(11, \text{category}) \mapsto \text{'Best Director'}$
 - $(12, \text{category}) \mapsto \text{'Best Supporting Actor'}$
 - $(13, \text{category}) \mapsto \text{'Best Movie'}$
 - $(16, \text{role}) \mapsto \text{'Vito Corleone'}$
 - $(18, \text{role}) \mapsto \text{'Michael Corleone'}$
 - $(19, \text{role}) \mapsto \text{'Michael Corleone'}$

Exercise 3 - Objectified Path Property Graph.

For the property graph and its formal definition from the first two exercises,

- Develop a meaningful extension to an *Objectified Path Property Graph*.
- Extend the formal definition from the Property Graph to an OPPG $G = (V, E, P, \eta, \delta, \lambda, \nu)$.



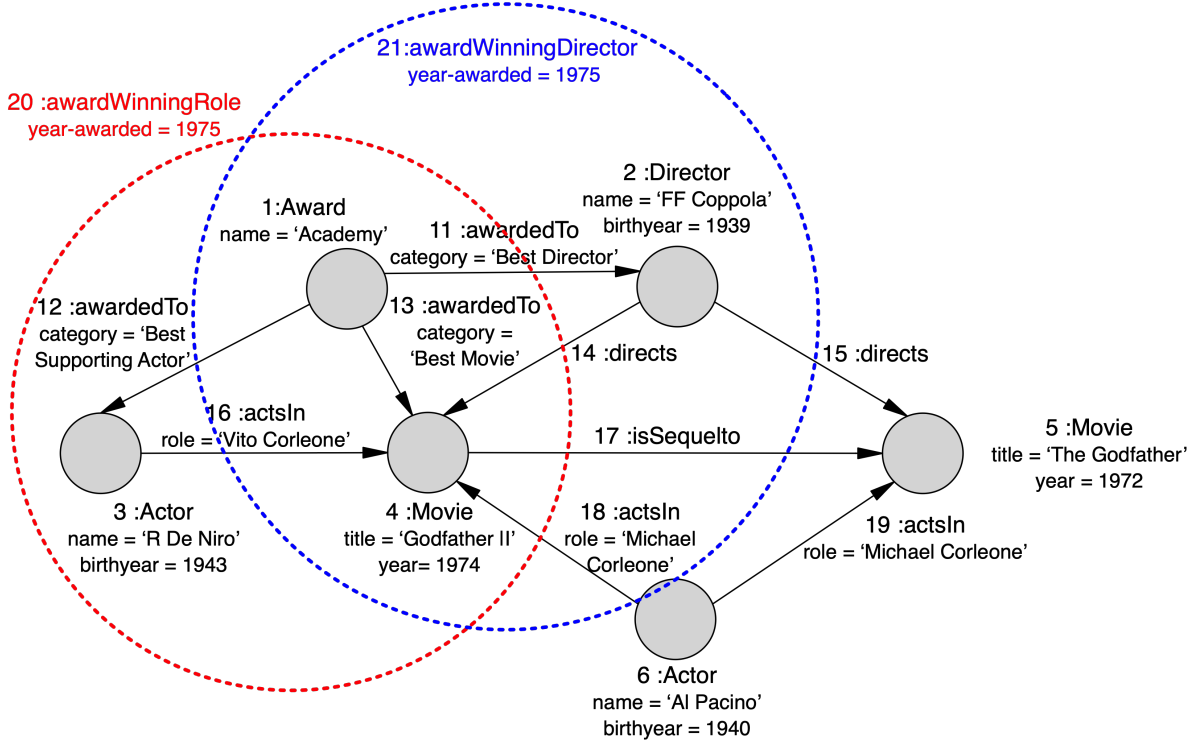
The following definitions need to be added to the formal property graph from the previous exercise to obtain a formal definition for the objectified path property graph above:

- $P = \{20\}$
- $\delta(20) = [13, 17]$
- $\lambda(20) = \text{'AwardWinningSequel'}$
- $\nu(20, \text{category}) = \text{'Academy'}$

Exercise 4 - Objectified Subgraph Property Graph.

For the property graph and its formal definition from the first two exercises,

- Develop a meaningful extension to an *Objectified Subgraph Property Graph*.
- Extend the formal definition from the Property Graph to an OSPG $G = (V, E, G, \eta, \gamma, \lambda, \nu)$.



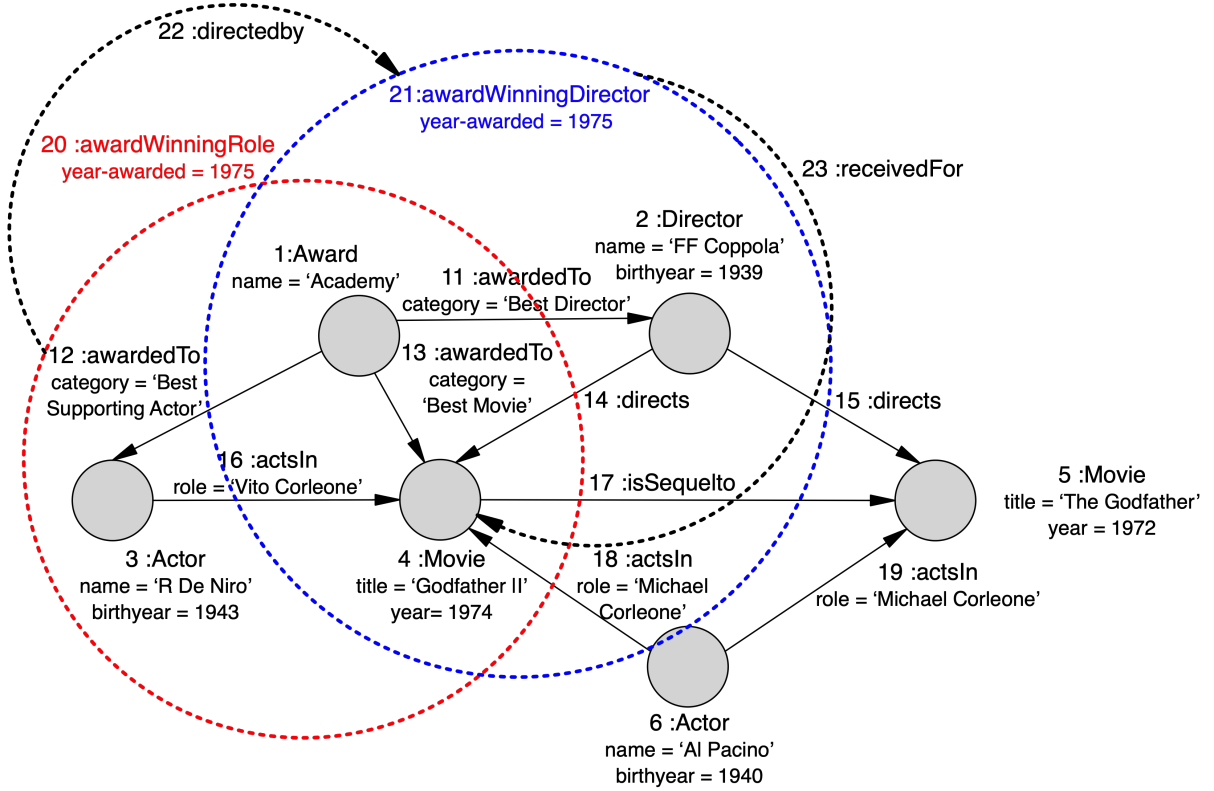
The following definitions need to be added to the formal property graph from the second exercise to obtain a formal definition for the objectified subgraph property graph above:

- $G = \{20, 21\}$
- γ :
 - $20 \mapsto (\{1, 3, 4\}, \{12, 13, 16\})$
 - $21 \mapsto (\{1, 2, 4\}, \{11, 13, 14\})$
- λ :
 - $20 \mapsto \text{:awardWinningRole}$
 - $21 \mapsto \text{:awardWinningDirector}$
- ν :
 - $(20, \text{year-awarded}) \mapsto 1975$
 - $(21, \text{year-awarded}) \mapsto 1975$

Exercise 5 - Hypervertex Property Graph.

For the subgraph property graph from the previous exercise,

- Develop a meaningful extension to a *Hypervertex Property Graph*.
- Extend the formal definition from the Objectified Subgraph Property Graph to an HVPG $G = (V, E, \eta, \gamma, \lambda, \nu)$.



The property graph that constitutes one solution for the first exercise is formally defined as an extension of the formal property graph model from the second exercise as follows:

- $V = \{1, 2, 3, 4, 5, 6, 20, 21\}$
- $E = \{11, 12, 13, 14, 15, 16, 17, 18, 19, 22, 23\}$
- η :
 - ...
 - $22 \mapsto (20, 21)$
 - $23 \mapsto (21, 4)$
- γ :
 - $1 \mapsto (\{1\}, \emptyset)$
 - ...
 - $20 \mapsto (\{1, 3, 4\}, \{12, 13, 16\})$
 - $21 \mapsto (\{1, 2, 4\}, \{11, 13, 14\})$
- λ :
 - ...
 - $20 \mapsto \text{:awardWinningRole}$

- $21 \mapsto \text{:awardWinningDirector}$
 - $22 \mapsto \text{:directedBy}$
 - $23 \mapsto \text{:receivedFor}$
- $\nu :$
- $\dots$
 - $(20, \text{year-awarded}) \mapsto 1975$
 - $(21, \text{year-awarded}) \mapsto 1975$